# MILITARY SPECIFICATION

# WARHEAD SECTION,

# LANCE

# GUIDED MISSILE,

# PRACTICE: XM198E1

## 1. SCOPE

1.1 This specification covers the metal parts, Loading, Assembling and Packing for one type of Warhead designated as Warhead Section: XM198E1.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-2550 | - | Ammunition and Special Weapons, General Specification for |
| MIL-T-5021 | - | Tests, Aircraft Welding Operator's Certification |
| MIL-I-6866 | - | Inspection, Penetrant Method of |
| MIL-R-11470 | - | Radiographic Inspection: Qualification of Equipment Operators and Procedures |
| MIL-I-45607 | - | Inspection Equipment, Acquisition, Maintenance and Disposition of |
| MIL-S-45743C | - | Soldering, Manual Type, High Reliability, Electrical Connections for Missile Systems, Procedures for |

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STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105) |
| MIL-STD-109 | - | Quality Assurance Terms and Definitions |
| MIL-STD-202 | - | Test Methods for Electronic and Electrical Component Parts |
| MIL-STD-1168 | - | Lot Numbering of Ammunition |
| MIL-STD-1235 | - | Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

| | | |
|---|---|---|
| 9267722 | - | Warhead Section, G.M., Practice: XM198E1 |

PUBLICATIONS

ARMY

| | | |
|---|---|---|
| ET-9267722 | - | Equipment Tabulation |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall appply.

CODE OF FEDERAL REGULATIONS

| | | |
|---|---|---|
| Title 49 | - | Transportation, Parts 0-190 |



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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and publications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) 9267722, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Radiographic examination of explosive assemblies.- The Double Lead Cup assemblies shall be radiographically examined for maximum gap between the detonating cord and Lead Cup assembly.

3.4 Soldering.- Soldering of all electrical joints and connections shall be in accordance with specification MIL-S-45743-C.

3.5 Functioning of Nose Assembly.- The Nose Assembly when disengaged shall be capable of being moved forward a distance of 0 inches minimum and rotated 105 degrees minimum clockwise and returned to the original position and be securely re-engaged without any motion stoppage due to interference between parts.

3.6 Welding.-All welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of deposit representative of good welding practice. The process shall be such as to prevent warpage beyond dimensional requirements. Sag of material in the area adjacent to the weld shall not be more than .010 below the unsagged surface. Cracks shall not exceed 1/2 inch in length and porous areas or linear porosity (spaced less than 1/16 inch apart) shall not exceed 1/2 inch in length. The total length of all defects in any weld shall not exceed 2 inches. Blow outs may be repaired by hand welding provided they do not exceed 1/4 inch in width and 1 inch in length. No more than 3 such areas may be repaired in any section. Only welders qualified in accordance with specification MIL-T-5021 shall be permitted to weld.

3.7 Electrical checkout of Cable, Tactical.- The Tactical Cable shall meet the continuity and insulation resistance requirements of dwg. 9271184.

3.8 Electrical checkout of assembled XM198E1 Warhead.- With the XM811E5 fuze disconnected, the warhead shall meet the insulation resistance requirements of dwg. 9271184. With the fuze, XM811E5 assembled



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to the warhead, the warhead shall meet the continuity requirements of Dwg. 9271184. NOTE: Do Continuity Test last. If the fuze is removed from the warhead, repeat the Continuity Test.

3.9 Workmanship.-All parts and assemblies shall be fabricated, finished, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.10 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (see 6.4).

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this Purchase Description.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual



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documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

4.2.1 Proving ground test.- When the provisions of AMCR 700-34, which require initial Production Testing apply, such testing shall be accomplished in accordance with 6.5.

4.2.2 Rejection.- If any assembly or component fails to comply with any of the applicable requirements, the lot shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action or submit a new lot quantity or portion thereof.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by



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the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 This paragraph applies to all components not listed and listed in 4.3.2.2 through 4.3.2.22 which require a protective finish (see applicable drawings) (See 3.9).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: None defined | | | |
| Minor: | 100% Inspection | | |
| 201. | Protective finish missing or inadequate | Visual | 01001 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 01002 |

4.3.2.2 Pin, Alignment (See Dwg. 8877425).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |



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| | | | |
|---|---|---|---|
| Major: | 100% Inspection | | |
| 101. | Concentricity of thread with body diameter | Gage | 02001 |
| Major: | 100% Inspection | | |
| 102. | Diameter of Body | Gage | 02002 |
| Minor: | AQL 1.00 Percent | | |
| 201. | Undercut missing | Visual | 02003 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 02004 |

4.3.2.3 Pin, Shear (See Dwg 8877426).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Concentricity of thread with body diameters | Gage | 03001 |
| Major: | 100% Inspection | | |
| 102. | Diameter of body | Gage | 03002 |
| Minor: | 100% Inspection | | |
| 201. | Undercut missing | Visual | 03003 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 03004 |

4.3.2.4 Sleeve (See Dwg. 9208748).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None Defined. | | |
| Major: | 100% Inspection | | |
| 101. | Thickness of Flange | Gage | 04001 |
| Minor: | 100% Inspection | | |
| 201. | Diameter of Flange (Min) | Gage | 04002 |
| 202. | Inside diameter (Max) | Gage | 04003 |
| 203. | Evidence of poor workmanship (See 3.9) | Visual | 04004 |



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4.3.2.5 Connector (See Dwg. 9267739).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Pitch diameter of thread | Gage | 05001 |
| 102. | Length to shoulder | Gage | 05002 |
| Minor: | 100% Inspection | | |
| 201. | Presence of chamfer at thread end | Visual | 05003 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 05004 |

4.3.2.6 Ring, Fuze (9267729).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Location of large tapped holes | Gage | 06001 |
| 102. | Location of fuze mounting holes | Gage | 06002 |
| 103. | Concentricity of largest outside diameter with smallest outside diameter | Gage | 06003 |
| 104. | Location of cable connector mounting holes | Gage | 06004 |
| 105. | Location of containing block mounting holes | Gage | 06005 |
| 106. | Depth of connector mounting surface | Gage | 06006 |
| Minor: | 100% Inspection | | |
| 201. | Pitch diameter of fuze mounting holes | Gage | 06007 |
| 202. | Pitch diameter of connector mounting holes | Gage | 06008 |
| 203. | Pitch diameter of large tapped holes | Gage | 06009 |
| 204. | Evidence of poor workmanship (See 3.9) | Visual | 06010 |



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4.3.2.7 Ring, Center (See Dwg. 9208515) -

| Categories | Defects | Method of Inspection | Code |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Concentricity of largest out side diameter with smallest outside diameter | Gage | 07001 |
| Minor: | 100% Inspection | | |
| 201. | Evidence of poor workmanship (See 3.9) | Visual | 07002 |

4.3.2.8 Ring, Aft (See Dwg. 9208266).-

| Categories | Defects | Method of Inspection | Code |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Location of shear pin holes | Gage | 08001 |
| 102. | Location of alignment pin holes | Gage | 08002 |
| Major: | 100% Inspection | | |
| 103. | Concentricity of shear pin threaded hole(s) | Gage | 08003 |
| 104. | Concentricity of alignment pin threaded hole(s) | Gage | 08004 |
| 105. | Depth of aft seal groove | Gage | 08005 |
| 106. | Outside diameter | Gage | 08006 |
| 107. | Tapped holes broken through | Visual | 08007 |
| Minor: | 100% Inspection | | |
| 201. | Pitch diameter of shear pin hole(s) | Gage | 08008 |
| 202. | Pitch diameter of alignment pin hole(s) | Gage | 08009 |
| 203. | Diameter of shear pin hole(s) | Gage | 08010 |
| 204. | Diameter of alignment pin hole(s) | Gage | 08011 |
| 205. | Concentricity of aft seal groove with outside diameter | Gage | 08012 |
| 206. | Width of aft seal groove | Gage | 08013 |



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|---|---|---|---|
| 207. | Inside diameter at aft seal groove | Gage | 08014 |
| 208. | Concentricity of inside diameter at aft seal groove | Gage | 08015 |
| 209. | Evidence of poor workmanship (See 3.9) | Gage | 08016 |

4.3.2.9 Aft Ballast Support Assembly (See Dwg. 9271189).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Runout of forward plate with aft ballast ring | Gage | 09001 |
| 102. | Parallelism of forward plate with aft ring | Gage | 09002 |
| 103. | Welding improper or incorrect (See 3.6) | Visual | 09003 |
| Minor: | 100% Inspection | | |
| 201. | Outside diameter | Gage | 09004 |
| 202. | Orientation of aft plate incorrect | Gage | 09005 |
| 203. | Evidence of poor workmanship (See 3.9) | Visual | 09006 |

4.3.2.10 Skin, Forward (See Dwg. 9208259).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Perpendicularity of open ends with inside diameters | Gage | 10001 |
| 102. | Contour of skin incorrect | Gage | 10002 |
| 103. | Welding improper or incorrect (when applicable)(See 3.6) | Gage/Visual | 10003 |
| Minor: | 100% Inspection | | |
| 201. | Thickness before forming | Gage | 10004 |
| 202. | Length | Gage | 10005 |
| 203. | Evidence of poor workmanship (See 3.9) | Visual | 10006 |



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4.3.2.11 Skin, Aft (See Dwg. 9208511).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Perpendicularity of forward and aft end with inside diameters | Gage | 11001 |
| 102. | Skin contour incorrect | Gage | 11002 |
| 103. | Welding improper or incorrect (When applicable) | Gage/Visual | 11003 |
| Minor: | 100% Inspection | | |
| 201. | Length | Gage | 11004 |
| 202. | Thickness (before forming) | Gage | 11005 |
| 203. | Evidence of poor workmanship (See 3.8) | Visual | 11006 |

4.3.2.12 Skin, Nose (See Dwg. 9208291).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Contour of skin incorrect | Gage | 12001 |
| 102. | Perpendicularity of open end | Gage | 12002 |
| Minor: | 100% Inspection | | |
| 201. | Length | Gage | 12003 |
| 202. | Thickness | Gage | 12004 |
| 203. | Evidence of poor workmanship (See 3.9) | Visual | 12005 |

4.3.2.13 Nose Assembly (See Dwg. 9208608).-

| Categories | Defects | Method of Inspection | Code |
|---|---|---|---|
| Critical: | None defined. | | |



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Major: None defined.

| Minor: | 100% Inspection | | |
|---|---|---|---|
| 201. | Any component missing, damaged or improperly assembled | Visual | 13001 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 13002 |

4.3.2.14 Ring Assembly (See Dwg. 9268135).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | 100% Inspection | | |
| 101. | Soldering incomplete | Visual | 14001 |
| Minor: | 100% Inspection | | |
| 201. | Thickness of ring (See Dwg. 9268146) | Gage | 14002 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 14003 |

4.3.2.15 Well, Charge (See 9268134).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Pitch diameter of threaded holes in flange | Gage | 15001 |
| 102. | Pitch diameter of ring assembly holes | Gage | 15002 |
| 103. | Pitch diameter of center threaded hole | Gage | 15003 |
| 104. | Depth of small countervore | Gage | 15004 |
| Minor: | 100% Inspection | | |
| 201. | Diameter of spotting charge well | Gage | 15005 |
| 202. | Length of spotting charge well | Gage | 15006 |
| 203. | Diameter of small counterbore | Gage | 15007 |
| 204. | Evidence of poor workmanship (See 3.9) | Visual | 15008 |



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4.3.2.16 Lead Cup Assembly (See Dwg. 9208255).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Density of PETN | Balance | 16001 |
| Minor: | 100% Inspection | | |
| 201. | Evidence of poor workmanship (See 3.9) | Visual | 16002 |

4.3.2.17 Double Lead Cup Assembly (See Dwg. 9208519).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Cup damaged or explosive is exposed | Visual | 17001 |
| Major: | 100% Inspection | | |
| 101. | Crimp(s) missing or incomplete | Visual | 17002 |
| Major: | 100% Inspection | | |
| 102. | Length | Gage | 17003 |
| 103. | Diameter of lead cup assembly (max) | Gage | 17004 |
| Minor: | 100% Inspection | | |
| 201. | Evidence of poor workmanship (See 3.9) | Visual | 17005 |

4.3.2.18 Cable Tactical (See Dwg. 9271184).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Wiring improper or incorrect | Visual | 18001 |
| 102. | Soldering improper, inadequate or omitted | Manual/Visual | 18002 |



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| 103. | Any component missing or damaged | Visual | 18003 |
| 104. | Potting missing, improper or inadequate | Visual | 18004 |
| 105. | Adhesive missing, or inadequate | Visual | 18005 |
| 106. | Wire insulation, frayed or charred | Visual | 18006 |
| 107. | Wires spliced | Visual | 18007 |
| 108. | Ground wires omitted, improper or inadequately installed | Manual/Visual | 18008 |
| Minor: | 100% Inspection | | |
| 201. | Overall Length | Gage | 18009 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 18010 |

4.3.2.19 XM198E1 Warhead Section, G.M., Practice (See Dwg. 9267722).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Detonating cord(s) or lead cup(s) damaged or explosive is exposed | Visual | 19001 |
| 2. | Red flag visible through window of fuze | Visual | 19002 |
| 3. | Setting of fuze incorrect | Visual | 19003 |
| 4. | Green flag not visible through window | Visual | 19004 |
| Major: | 100% Inspection | | |
| 101. | Improper position of lead cups in containing block | Gage | 19005 |
| 102. | Weight incorrect | Scale | 19006 |
| 103. | Torque of fuze mounting screw(s) incorrect | Gage | 19007 |
| 104. | Fuze mounting screw seals missing or broken | Visual | 19008 |
| 105. | Operation of nose assembly improper (See 3.5) | Manual | 19009 |
| 106. | Any component missing or damaged | Visual | 19010 |



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|---|---|---|---|
| 107. | Orientation of J1 connector improper | Visual | 19011 |
| 108. | Skin gaps excessive | Gage | 19012 |
| 109. | Skin surface mismatch excessive | Gage | 19013 |
| 110. | Carriers not secured to skin over entire length | Visual | 19014 |
| 111. | Aft seal missing | Visual | 19015 |
| 112. | Sleeve crimped to double lead cup assembly | Visual | 19016 |
| 113. | Explosive cross-tie secure | Visual/Manual | 19017 |
| 114. | Fuze gasket missing or improperly assembled | Visual | 19018 |
| 115. | Aluminum tape missing or improper | Visual | 19019 |
| 116. | Tactical cable not properly dressed or assembled to Warhead Section | Visual | 19020 |
| Minor: | 100% Inspection | | |
| 201. | Protrusion of shear pin(s) incorrect | Gage | 19021 |
| 202. | Protrusion of alignment pin(s) incorrect | Gage | 19022 |
| 203. | Connector spring pin above flush too far below flush | Gage | 19023 |
| 204. | Distance from aft skin to aft ring incorrect | Gage | 19024 |
| 205. | Distance from aft skin to center ring incorrect | Gage | 19025 |
| 206. | Distance from forward skin to fuze ring incorrect | Gage | 19026 |
| 207. | Double lead cup assemblies not seated in carrier | Visual | 19027 |
| 208. | Evidence of poor workmanship (See 3.9) | Visual | 19028 |

4.3.2.20 XM198E1 Warhead Section, G.M., Practice, After painting and marking (See Dwg. 9267722).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | None defined. | | |



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|---|---|---|---|
| Minor: | 100% Inspection | | |
| 201. | Marking missing, incorrect or unidentifiable | Visual | 20001 |
| 202. | Evidence of poor workmanship (See 3.9) | Visual | 20002 |

4.3.2.21 XM198E1 Warhead Section, G.M., Practice, After assembling into container and prior to assembling cover (See Dwg. 9211190).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Torque of Warhead "T" bolts incorrect | Gage | 21001 |
| Minor: | 100% Inspection | | |
| 201. | Evidence of poor workmanship (See 3.9) | Visual | 21002 |

Minor: None defined.

4.3.2.22 XM198E1 Warhead Section, G.M., Practice, Container, after assembly of cover (See Dwg. 9211190).-

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | 100% Inspection | | |
| 101. | Any bolt not fastened properly | Visual | 22001 |
| 102. | Seal missing or broken | Visual | 22002 |
| Minor: | 100% Inspection | | |
| 201. | Marking misleading, unidentifiable or incorrect | Visual | 22003 |
| 202. | Evidence of poor workmsnship (See 3.9) | Visual | 22004 |

4.3.3.1 Dimensional control of cast parts.- In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally



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inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.3.3.2 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.3 Pull test of crimp of explosive assemblies (See applicable dwg), Major defect.- Each explosive assembly listed below shall be subjected to this test. If any assembly fails to comply with the specified requirement, that lot assembly shall be rejected. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4. Code numbers for this



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test are as follows:

| Assembly | Code No. |
|---|---|
| *Double Lead Cup Assembly (9208519) | 41001 |
| Sleeve (9208748) | 41002 |

*After completion of this test, these assemblies shall meet the requirements of 3.3 when tested as specified in 4.4.2.

4.3.3.4 Radiographic examination of explosive assemblies and double lead cup assembly (9208519) (See 3.3), Code No. 42001.- This test shall be performed 100 percent. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.5 Die Penetrant inspection of Aft Ring (See Dwg. 9208266), and Center Ring (See dwg. 9208515), Major defect, Code No. 43001.- Each assembly shall be subjected to this test. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in MIL-I-6866, Type II, Method A, using equipment specified therein.

4.3.3.6 Electrical continuity of Tactical Cable before assembly to warhead section (See Dwg. 9271184), Major defect, Code No. 44001.- Each Tactical Cable shall be subjected to this test. Any Tactical Cable which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

4.3.3.7 Insulation resistance of Tactical Cable before assembly to warhead section (See dwg. 9271184), Major defect, Code No. 44002.- Each Tactical Cable shall be subjected to this test. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.8 Functioning of Nose Assembly (See Dwg. 9267722) (3.5), Major Defects, Code No. 46001.- This test shall be performed 100 percent on every assembled warhead. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. These tests shall be performed as specified in the paragraphs indicated below, using equipment in accordance with 4.3.3.



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4.3.3.9 Physical characteristics of assembled XM198E1 Warhead (See Dwg. 9267722), Major defect.-These tests shall be performed 100 percent. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. These tests shall be performed as specified in the paragraphs indicated below, using equipment in accordance with 4.3.3. Code Numbers for these tests are as follows:

| Characteristics | Code No. | Test Para. |
|---|---|---|
| Longitudinal Center of Gravity | 47001 | 4.4.5.1 |
| Radial Center of Gravity | 47002 | 4.4.5.2 |
| Weight of assembled Warhead | 47003 | 4.4.5.3 |

4.3.3.10 Electrical checkout of assembled Warhead (See 3.8), Major defect, Code No. 48001.- Each assembled warhead shall be subjected to this test. Any assembly which fails to comply with 3.8 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9267722 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Pull test of crimp of explosive assembly.-The applicable explosive assembly shall be assembled to a fixture as specified on the equipment list and the required force shall be applied. This is a destructive test. Assemblies so tested shall not be returned to the lot.



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4.4.2 Radiographic examination of explosive assemblies.-The explosive assemblies shall be positioned on a fixture as specified on the equipment list. Each assembly shall be identified to its position on the radiograph and the exposure made. Examinations of the radiographs shall be made for compliance with the applicable cap requirement. Radiographic equipment, operators and procedures shall be qualified in accordance with specification MIL-R-11470. This is a non-destructive test. Parts so tested may be returned to the lot if acceptable.

4.4.3 Electrical continuity of Tactical Cable before assembly to Warhead Section.- The Tactical Cable shall be assembled to a fixture as specified on the equipment list and tested for compliance with the specified requirement. This is a non-destructive test.

4.4.4 Insulation resistance of Tactical Cable before assembly to Warhead Section.- The Tactical Cable shall be assembled to a fixture as specified on the equipment list and tested for compliance with the specified requirement. This is a non-destructive test. Assemblies so tested may be returned to the lot if acceptable.

4.4.5 Physical characteristics of assemblies XM198E1 Warhead.-

4.4.5.1 Longitudinal center of gravity.- The assembled warhead shall be assembled to a fixture as specified on the equipment list and tested for compliance with the specified requirement. This is a non-destructive test. Assemblies so tested may be returned to the lot if acceptable.

4.4.5.2 Radial center of gravity.- The assembled warhead list and tested for compliance with the specified requirement. This is a non-destructive test. Assemblies so tested may be returned to the lot if acceptable.

4.4.5.3 Weight of Assembled Warhead.- The assembled warhead shall be assembled to a fixture as specified on the equipment list and tested for compliance with the specified requirement. This is a non-destructive test. Assemblies so tested may be returned to the lot if acceptable.

4.4.6 Electrical checkout of assembled Warhead (9267722).- The equipment as specified on the Equipment List shall be assembled to the warhead and checkout for compliance with the specified continuity requirement shall be performed. The insulation resistance shall be conducted with the XM811 Fuze disconnected and with the warhead grounded. This is a non-destructive test. Warheads so tested may be returned to the lot if acceptable.

NOTE: Do continuity test last, if the fuze is removed from the warhead after continuity test and re-installed, the continuity test must be repeated.



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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.-The XM198 Warhead shall be preserved and packaged in accordance with the applicable Dwg.

5.2 Packing.-

5.2.1 Level A.-The XM198 Warhead, preserved and packaged as specified in 5.1.1, shall be packed in accordance with the applicable Dwg.

5.3 Marking.-Marking shall be in accordance with the applicable dwgs and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples. (when spec requires first article sampling).

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.



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6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on



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the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subjected to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional marking of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall froward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.5 Proving ground test.- Initial production testing shall be conducted in accordance with U.S. Army Test and Evaluation Command Initial Production Test Plan No. _____.

Custodian:
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Preparing Activity:
Army-MU

Project Number 1340-A315



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